

DPT Exercises

Exercise sheet 2

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Exercise 1. Let $(X, \mathcal{A}_X)$ and $(Y, \mathcal{A}_Y)$ be measurable spaces, $f : X \rightarrow Y$ a map, and $\mathcal{E}_Y \subseteq \mathcal{A}_Y$ arbitrary. Show that

$$\sigma(f^{-1}(\mathcal{E}_Y)) = f^{-1}(\sigma(\mathcal{E}_Y)).$$

Further, show that

$$f \text{ is } \mathcal{A}_X\text{-}\sigma(\mathcal{E}_Y)\text{-measurable} \iff f^{-1}(E') \in \mathcal{A}_X \text{ for all } E' \in \mathcal{E}_Y.$$

With that, finally show that for $\sigma(\mathcal{E}_Y) = \mathcal{A}_Y$ it holds that

$$f \text{ is } \mathcal{A}_X\text{-}\mathcal{A}_Y\text{-measurable} \iff f^{-1}(\mathcal{E}_Y) \subseteq \mathcal{A}_X.$$

This means that to show that a function is measurable, it is enough to show measurability on a generator.

Solution: We show both inclusions separately:

“ $\subseteq$ ”: Since $f^{-1}(\mathcal{E}_Y) \subseteq f^{-1}(\sigma(\mathcal{E}_Y))$ and $f^{-1}(\sigma(\mathcal{E}_Y))$ is a σ -algebra, $\sigma(f^{-1}(\mathcal{E}_Y)) \subseteq f^{-1}(\sigma(\mathcal{E}_Y))$.

“ $\supseteq$ ”: For the σ -algebra

$$\tilde{\mathcal{A}} = \{A' \in \sigma(\mathcal{E}_Y) \mid f^{-1}(A') \in \sigma(f^{-1}(\mathcal{E}_Y))\} = f_*(\sigma(f^{-1}(\mathcal{E}_Y)) \cap \sigma(\mathcal{E}_Y)) \subseteq \sigma(\mathcal{E}_Y)$$

we obtain $\tilde{\mathcal{A}} = \sigma(\mathcal{E}_Y)$ and thus $f^{-1}(\sigma(\mathcal{E}_Y)) = f^{-1}(\tilde{\mathcal{A}}) \subseteq \sigma(f^{-1}(\mathcal{E}_Y))$.

Exercise 2. Let $\Omega \neq \emptyset$ be a set, $A \subseteq \Omega$, $A \neq \emptyset$, and let $\mathcal{E} \subseteq \mathcal{P}(\Omega)$. Then it holds that

$$\sigma(\mathcal{E}|_A) = \sigma(\mathcal{E})|_A.$$

Solution: Consider the map $\iota_A : A \hookrightarrow X$ that is the natural injection, i.e., $\iota_A(a) = a$ for all $a \in A$. Then $\iota_A^{-1}(B) = A \cap B$ for all $B \subseteq X$. Therefore, we find

$$\begin{aligned} \sigma(\mathcal{E}|_A) &= \sigma(\{E \cap A \mid E \in \mathcal{E}\}) = \sigma(\{\iota_A^{-1}(E) \mid E \in \mathcal{E}\}) = \sigma(\iota_A^{-1}(\mathcal{E})) \stackrel{(*)}{=} \iota_A^{-1}(\sigma(\mathcal{E})) \\ &= \{A \cap B \mid B \in \sigma(\mathcal{E})\} = \sigma(\mathcal{E})|_A, \end{aligned}$$

where in $(*)$ we use the result from Exercise 1.

Exercise 3. Let $(p_n)_{n \in \mathbb{N}} \subset [0, 1]$ with a fixed $\lambda := \lim_{n \rightarrow \infty} n p_n \in \mathbb{R}_{>0}$. Then for all $\omega \in \mathbb{N}$ we have that

$$\lim_{n \rightarrow \infty} p_{\text{Bin}(n, p_n)}(\omega) = p_{\text{Poi}(\lambda)}(\omega).$$

Solution: Starting from $p_{\text{Bin}(n, p_n)}(\{k\}) = \binom{n}{k} p_n^k (1 - p_n)^{n-k}$, as we have

$$\binom{n}{k} = \frac{n!}{(n-k)! k!} = \frac{n \cdot (n-1) \cdots (n-k+1)}{k!},$$

first note that

$$\frac{\binom{n}{k}}{n^k/k!} \rightarrow 1 \quad \text{as } n \rightarrow \infty. \tag{1}$$

Moreover, as $np_n \rightarrow \lambda$, we furthermore have

$$(np_n)^k \rightarrow \lambda^k. \quad (2)$$

As $(p_n)_{n \in \mathbb{N}}$ is a null sequence, we in addition have

$$(1 - p_n)^k \rightarrow 1 \quad \text{as } n \rightarrow \infty. \quad (3)$$

Tying everything together gives:

$$\lim_{n \rightarrow \infty} \binom{n}{k} p_n^k (1 - p_n)^{n-k} = \lim_{n \rightarrow \infty} \frac{n^k}{k!} p_n^k (1 - p_n)^{n-k} \quad (\text{by (1)})$$

$$= \lim_{n \rightarrow \infty} \frac{(\lambda)^k}{k!} (1 - p_n)^{n-k} \quad (\text{by (2)})$$

$$= \frac{\lambda^k}{k!} \cdot \lim_{n \rightarrow \infty} (1 - p_n)^n = \frac{\lambda^k}{k!} \lim_{n \rightarrow \infty} \left(1 - \frac{np_n}{n}\right)^n \quad (\text{by (3)})$$

$$= \frac{\lambda^k}{k!} \cdot e^{-\lambda}. \quad (\text{by (*)})$$

where in (*) we used the approximation formula:

$$e^{-x} = \lim_{n \rightarrow \infty} \left(1 - \frac{x}{n}\right)^n.$$

Exercise 4. After a long game night, Agathe and Balthasar are standing alone at the subway station Marienplatz, where a train departs every 10 minutes. Unfortunately, the trains are so overcrowded that only one person can board each time. Every person on the platform has the same chance of boarding the train. Additionally, until the next train arrives, two new people arrive at the station. Determine the probability that Agathe will be able to board

- (i) the n -th train,
- (ii) at the latest by the n -th train,

where $n \in \mathbb{N}$.

Solution:

- (i) First, we note that the number of people at the station increases by one with each train: one person boards, but two new people arrive. Since initially two people are at the station, there are exactly $n + 1$ people when the n -th train arrives.

When the i -th train arrives, Agathe can board with probability $\frac{1}{i+1}$. The probability that she doesn't board any of the first $n - 1$ trains is therefore:

$$\prod_{i=1}^{n-1} \left(1 - \frac{1}{i+1}\right) = \prod_{i=1}^{n-1} \frac{i}{i+1} = \frac{1}{2} \cdot \frac{2}{3} \cdot \dots \cdot \frac{n-2}{n-1} \cdot \frac{n-1}{n} = \frac{1}{n}.$$

Finally, Agathe must be the one to get the free spot on the n -th train, which occurs with probability $\frac{1}{n+1}$. Hence, the total probability is:

$$\frac{1}{n} \cdot \frac{1}{n+1} = \frac{1}{n(n+1)}.$$

- (ii) The events “Agathe boards at the latest with the n -th train” and “Agathe does not board any of the first n trains” are complementary.

From part (a), the latter occurs with probability $\frac{1}{n+1}$. Therefore, the probability that Agathe boards one of the first n trains is:

$$1 - \frac{1}{n+1} = \frac{n}{n+1}.$$